

Data Visualization using Tableau

Session 2

Agenda

- Creating Dashboards using Tableau

Introduction to Dashboard Creation

What is a Dashboard?

An interactive tool that visually tracks, analyzes, and displays key performance indicators, metrics, and data points.

Purpose of Dashboards in Tableau:

To provide actionable insights through aggregation and visualization of data.

Principles of Dashboard Design

- **Clarity and Simplicity:**

- The dashboard should convey insights at a glance. Use minimal design and avoid clutter.
- Ensure charts and data visualizations are easy to interpret, with labels, legends, and titles where necessary.

- **User-Centered Design:**

- Understand the audience and tailor the dashboard to their needs, ensuring relevant metrics are displayed.
- Design for usability by ensuring the dashboard layout is intuitive and navigable.

- **Data Prioritization:**

- Focus on key metrics (KPIs) that align with business goals.
- Arrange data hierarchically, with the most important insights at the top or center of the dashboard.

Principles of Dashboard Design

- **Consistency:**
 - Use consistent formatting, colors, and chart types across the dashboard to avoid confusion.
 - Follow a standard structure so users know where to find specific types of information.
- **Data Context and Comparisons:**
 - Provide context by using benchmarks, targets, or comparisons to give meaning to the data.
 - Include time-based trends or historical data to show progress or changes.
- **Interactivity and Flexibility:**
 - Allow users to filter, drill down, or hover over data points to get detailed information.
 - Provide customizable views so users can focus on their specific interests.

Principles of Dashboard Design

- **Performance and Speed:**
 - Ensure the dashboard loads quickly and is optimized for performance, especially with large datasets.
 - Avoid unnecessary or overly complex visualizations that slow down the dashboard.
- **Visual Balance and Aesthetics:**
 - Balance the visual elements with proper spacing and alignment for a clean, professional look.
 - Use color strategically to highlight important data points and trends.

Organize Your Dashboard

- **Define Objectives:**

- Clearly establish the purpose of the dashboard and what key insights it should provide.
- Identify the most important metrics (KPIs) that align with user goals and focus on delivering actionable data.

- **Logical Layout:**

- Use a structured layout that groups related information together. Typically, dashboards are arranged in the following way:
 - **Top Section:** Summary of key metrics or an overview.
 - **Middle Section:** Detailed insights, charts, and comparisons.
 - **Bottom Section:** Less critical information or further drill-down details.

Organize Your Dashboard

- **Prioritize Key Data:**
 - Place the most critical information at the top of the dashboard, as users typically focus here first.
 - Ensure that insights requiring immediate attention (e.g., red flags, anomalies) are prominent.
- **Group Similar Metrics:**
 - Organize metrics into sections or panels based on their relationships (e.g., sales data, customer behavior, operational metrics).
 - Group similar charts or data together to avoid switching contexts and help users focus on related data.
- **Use Consistent Visual Elements:**
 - Apply consistent fonts, colors, and chart types for similar data to maintain cohesion.
 - Ensure that the dashboard follows the same design rules for charts, filters, and labels, making it easier to follow.

Organize Your Dashboard

- **Apply Interactivity Thoughtfully:**
 - Incorporate filters, drill-downs, or hover-over details for interactive insights, without overwhelming the user.
 - Allow users to customize the view based on their needs while keeping the core layout intact.
- **Ensure Easy Navigation:**
 - For complex dashboards, use tabs, navigation bars, or buttons to organize different sections or types of data.
 - Provide a clear, easy-to-use navigation system that guides users through different views or reports.
- **Balance Visual Elements and Text:**
 - Use visualizations like charts and graphs to represent data concisely, while supplementing them with explanatory text where needed.
 - Avoid overwhelming users with too much text; let the data visuals do the storytelling.
- **Test and Iterate:**
 - Regularly test the dashboard's layout and structure with end users to ensure it meets their needs.
 - Collect feedback and improve the organization to make it more intuitive and responsive to evolving requirements.

Hierarchies

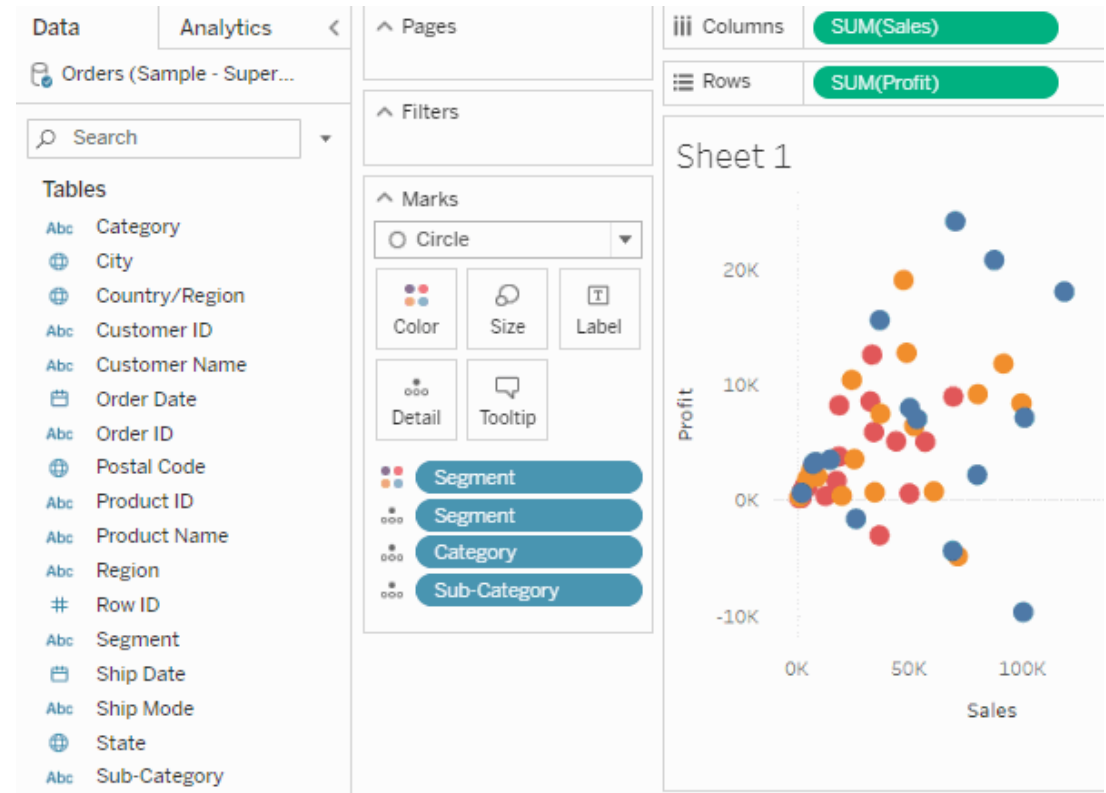
In Tableau, a hierarchy is a structured arrangement of data fields that defines the levels of data granularity and their relationships. It allows users to organize data fields into a tree-like structure where higher-level fields encompass more general information, and lower-level fields provide more detailed information.

1.Parent Level: The top level of the hierarchy that represents the most general category (e.g., Year in a time hierarchy, Country in a geographical hierarchy).

2.Child Levels: The subsequent levels that provide more detailed categories (e.g., Month and Day under Year, State and City under Country).

Hierarchies

- **Benefits of Using Hierarchies**
- **Improved Data Organization:** Helps in logically structuring data for easier analysis and visualization.
- **Enhanced Drill-down Capabilities:** Allows users to explore data at different levels of detail by drilling down or rolling up through the hierarchy.
- **Simplified Visualization:** Makes it easier to create nested charts and other visualizations that require hierarchical data.



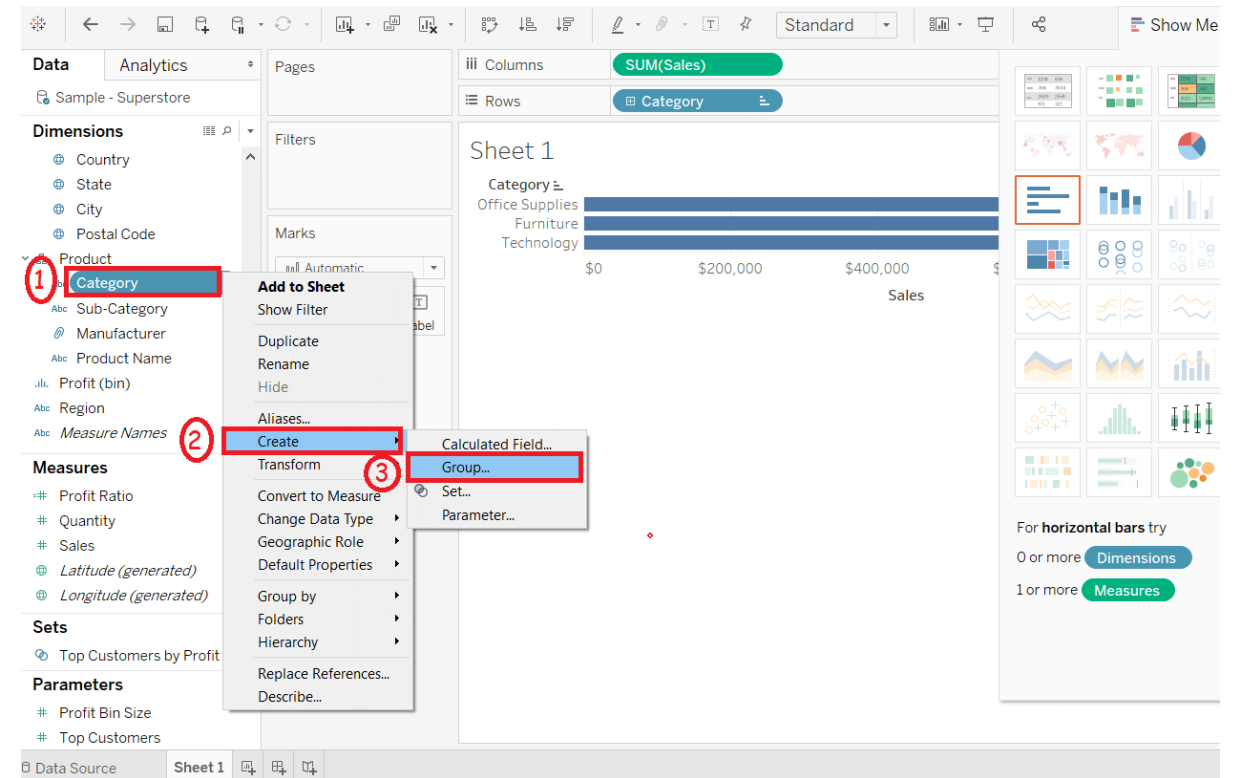
Grouping

Grouping in Tableau is a technique used to combine multiple dimension members into a single, higher-level category. This helps in simplifying complex datasets, making it easier to analyze and visualize data by aggregating similar items.

1. **Dimension Members:** Individual data points within a dimension that you want to group together.
2. **Group:** The new category created by combining selected dimension members.

Benefits of Using Grouping

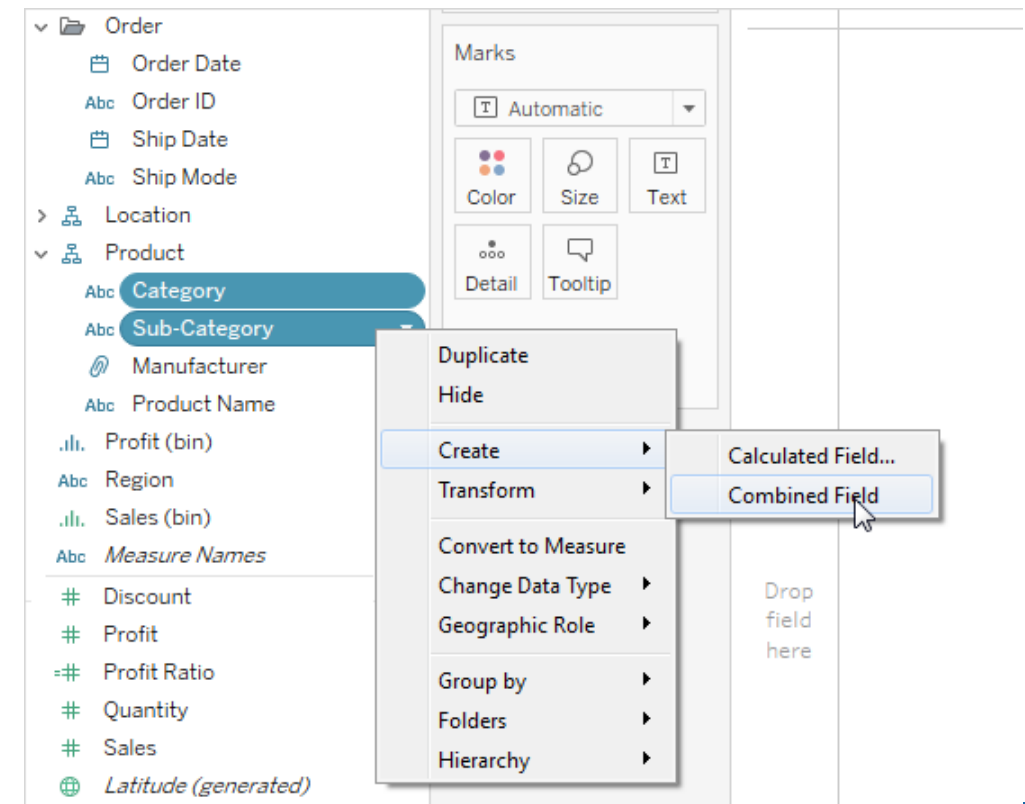
- **Simplified Analysis:** Reduces the number of categories in a dimension, making data easier to interpret.
- **Enhanced Visual Clarity:** Improves the readability of visualizations by aggregating similar items.
- **Custom Categorization:** Allows users to create custom categories that better align with their analysis needs.



Combined fields

A combined field in Tableau is a custom field created by merging two or more existing fields, typically dimensions, into a single field. This new field helps to form unique identifiers or multi-level categories, enhancing the granularity and specificity of data analysis.

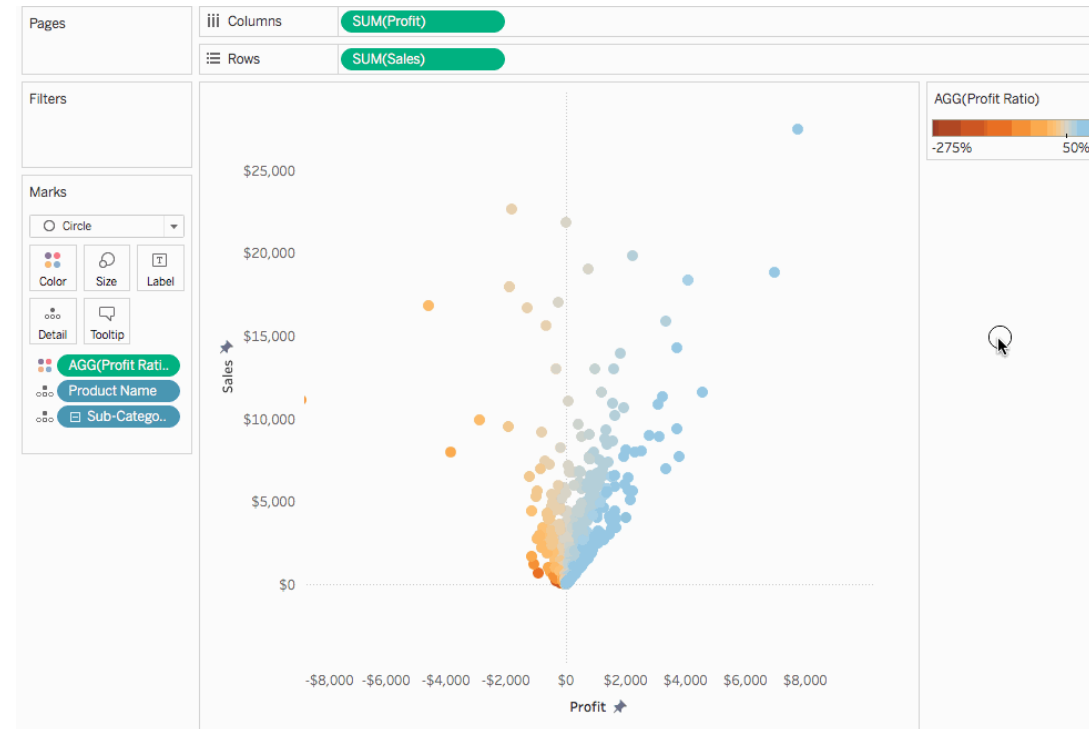
- **Merges Multiple Fields:** Combines two or more fields into a single field.
- **Unique Identifiers:** Useful for creating unique keys or identifiers.
- **Enhanced Granularity:** Allows for more detailed and specific categorization and analysis.
- **Concatenation:** Typically involves concatenating the values of the selected fields.



Sets

Sets in Tableau are custom fields that define a subset of data based on specific conditions. They enable users to group data dynamically and perform comparative analysis within visualizations. Sets can be based on a single dimension or a combination of multiple dimensions.

- **Dynamic Membership:** The members of a set can change dynamically based on the data and conditions defined.
- **Conditional Inclusion:** Sets are created based on conditions or criteria that determine which dimension members are included.
- **Comparative Analysis:** Sets allow for easy comparison between different subsets of data.



Bins

Bins in Tableau are containers that group data into specified intervals. They are particularly useful for creating histograms or analyzing the distribution of data. Bins are typically used with continuous measures to categorize data into discrete intervals.

- **Interval-based:** Bins divide data into equal-sized intervals.
- **Customizable:** Users can define the size of the bins.
- **Discrete:** Bins create discrete categories from continuous data.

Measures & Dimensions

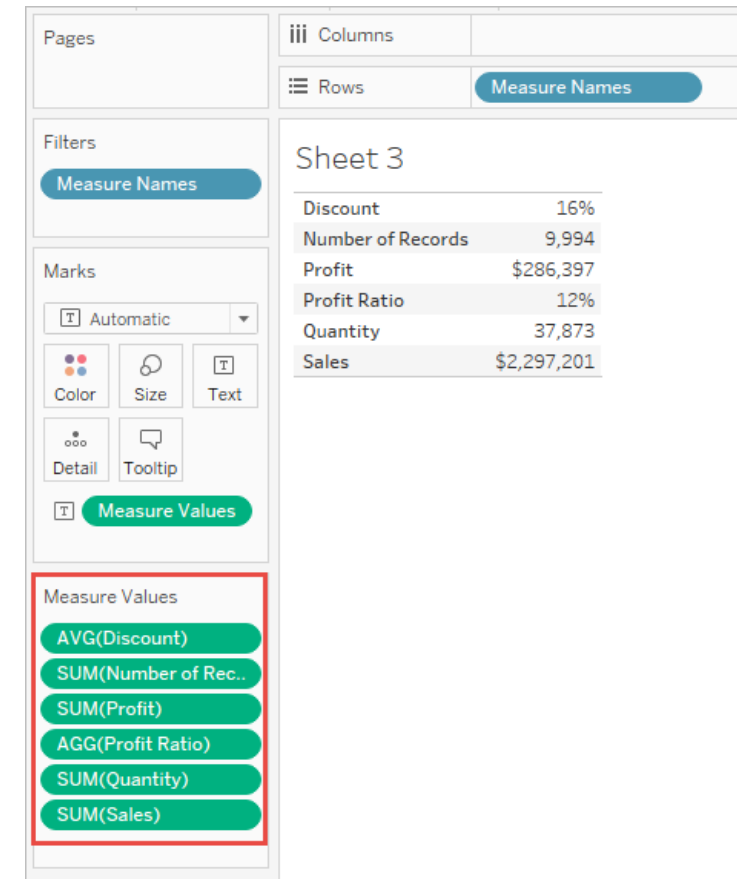
Measures

Measures are quantitative, numerical data that can be aggregated. They typically represent metrics such as sales, profit, quantity, and other numerical values that you want to analyze and visualize.

- **Quantitative:** Measures are always numerical.
- **Aggregative:** They can be summed, averaged, counted, or otherwise aggregated.
- **Context-Sensitive:** Their meaning can change based on the dimension they are paired with in an analysis.

Examples:

- Sales Revenue
- Profit Margin
- Quantity Sold
- Temperature Readings



The screenshot shows a Tableau interface with a table of measures and dimensions. The table is titled "Sheet 3" and has two columns: "Measure Names" and "Measure Values". The measures listed are Discount, Number of Records, Profit, Profit Ratio, Quantity, and Sales. The values are 16%, 9,994, \$286,397, 12%, 37,873, and \$2,297,201 respectively. The interface also shows a sidebar with filters, marks, and a list of measure values.

Measure Names	Measure Values
Discount	16%
Number of Records	9,994
Profit	\$286,397
Profit Ratio	12%
Quantity	37,873
Sales	\$2,297,201

Measures & Dimensions

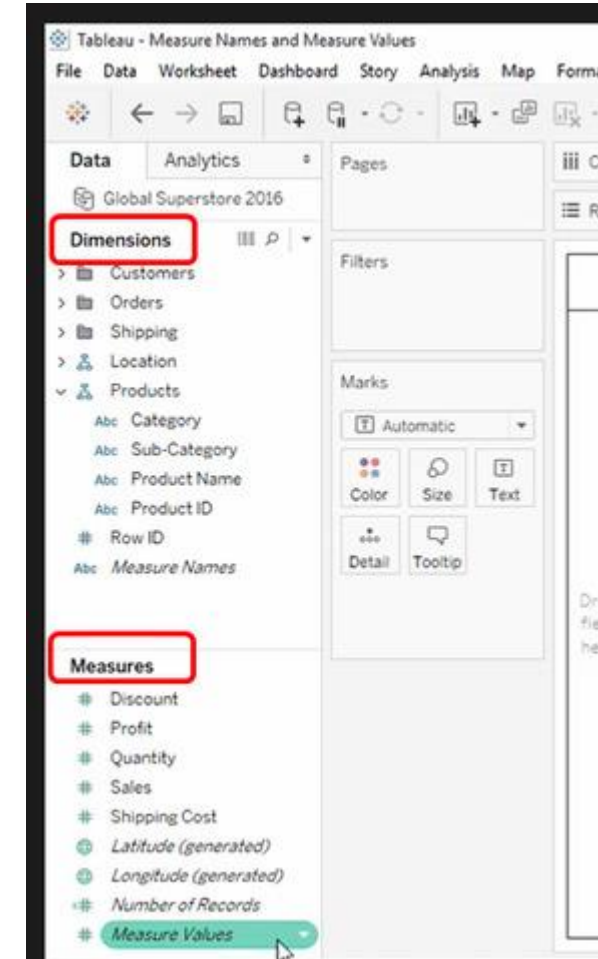
Dimensions

Dimensions are qualitative, categorical data that provide context for measures. They typically represent attributes

- **Qualitative:** Dimensions are typically non-numerical.
- **Contextual:** They provide context and structure to the data, helping to categorize, segment, and filter measures.
- **Descriptive:** Dimensions describe the "what", "who", "when", and "where" aspects of data.

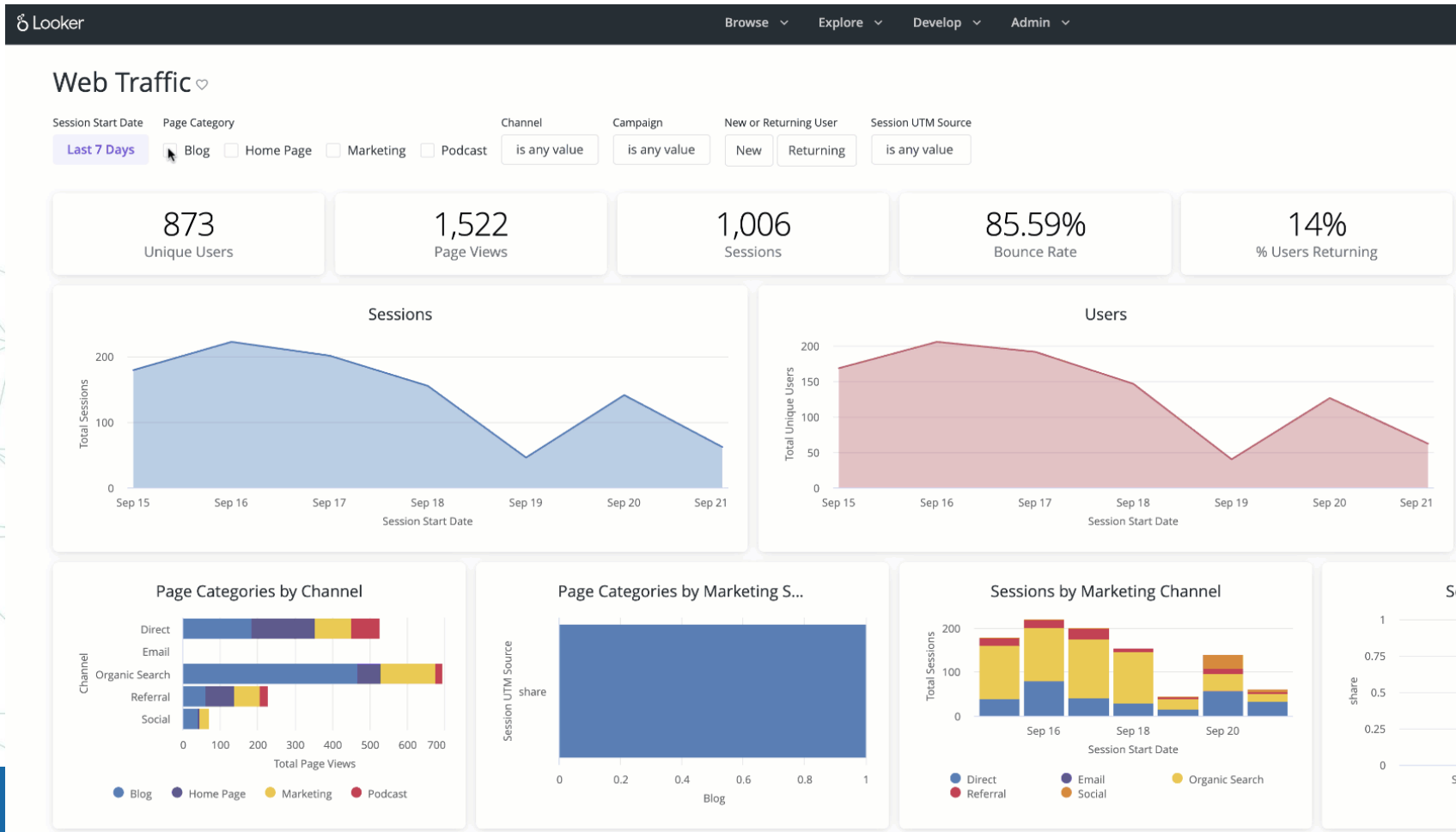
Examples:

- Product Name
- Customer ID
- Region
- Order Date



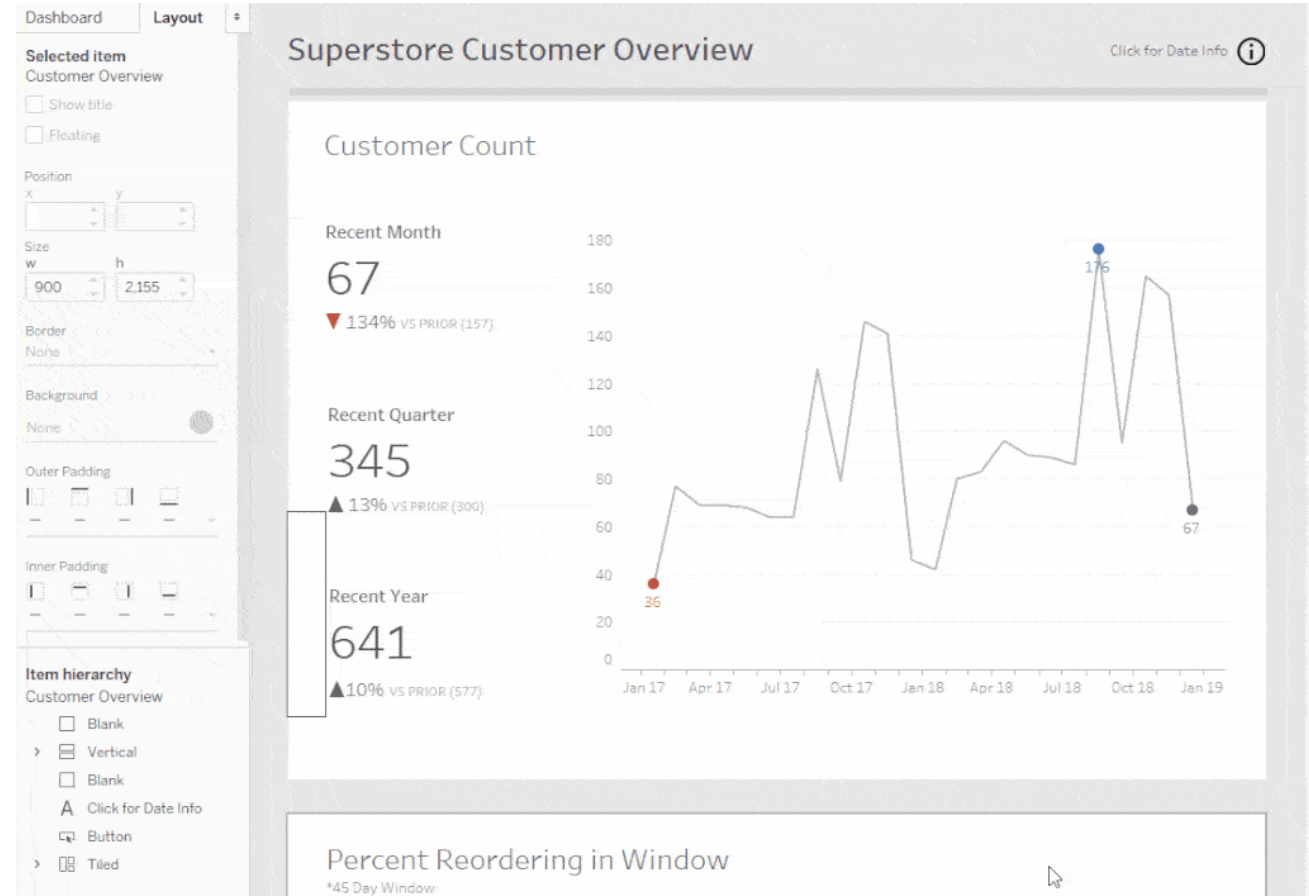
Combining Multiple Visualizations

- **Complementary Data:** Combine charts that add value to each other.
- **Cross-filtering:** Set up interactions between charts for deeper analysis.



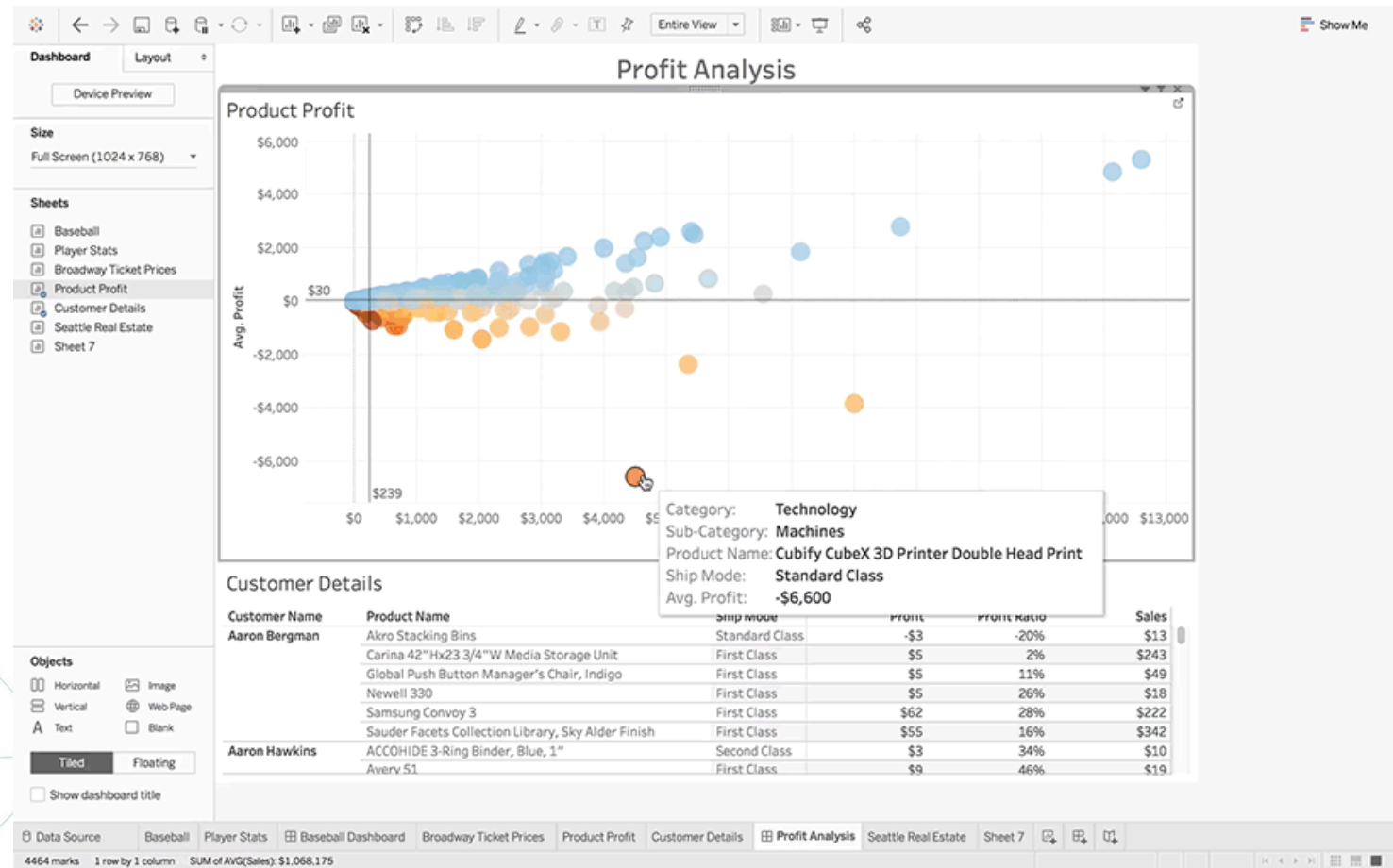
Layout Best Practices

- **Grid Layouts:** Use grids to align elements neatly.
- **Whitespace:** Utilize whitespace to prevent a feeling of overcrowding.



Interactive Elements in Dashboards

- **Filters and Actions:** Include interactive filters to allow viewers to drill down into specifics.
- **Tool Tips:** Use tooltips for additional context without cluttering the view.



Dashboard Performance Optimization

- **Data Refresh:** Optimize the frequency of data updates to balance freshness with performance.
- **Query Simplification:** Simplify data queries to speed up loading times.

Let's practice



Any Questions ?

Thank You